

Noel Kallingal

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Noel-Portfolio

EDUCATION

Vellore Institute of Technology - 8.26

B.Tech in Computer Science and Engineering

Vellore, Tamil Nadu

2023 – Current

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, JavaScript, Dart, HTML, CSS, SQL

Frameworks & Libraries: React.js, FastAPI, Flutter, Express.js, Django, Tailwind CSS, TensorFlow, MediaPipe, OpenCV, Scikit-learn, Pandas, NumPy

Databases: MySQL, PostgreSQL, ChromaDB, Firebase

Tools & Platforms: Git, GitHub, Vite, Ollama, PDFPlumber, REST APIs, Figma, Adobe Photoshop, Vercel, Render

WORK EXPERIENCE

Star Innovations - Web Development Intern

Jun 2024 – Jul 2024

Thrissur, Kerala

- Developed responsive websites using HTML, CSS, and JavaScript, ensuring cross-browser compatibility and optimal performance.
- Integrated a RESTful API to retrieve real-time weather data, enhancing user engagement with dynamic updates.
- Strengthened skills in API development, data handling, and frontend-backend integration for full-stack web development.

GramPro Business Services - App Development Intern

May 2025 – Jul 2025

Infopark Thrissur, Kerala

- Developed a Flutter-based project management app for tracking tasks, deadlines, and notifications.
- Created a responsive dashboard to display project status, milestones, and pending work.
- Handled backend databases and APIs with secure access, improving synchronization and team collaboration.

PROJECTS

Wearable Eye Fatigue Tracker for Professionals - Arduino, React, FastAPI, MediaPipe, TensorFlow

Vellore, Tamil Nadu

- Developed an AI-powered fatigue detection system using MediaPipe Face Mesh to extract blink rate, Eye Aspect Ratio (EAR), and EAR variance from real-time webcam input.
- Built and evaluated Random Forest and LSTM models to classify users into Focused, Neutral, and Fatigued states through a FastAPI backend.
- Integrated the AI pipeline with an Arduino wearable using optical sensors, IR sensors, and MPU6050 for continuous monitoring and real-time alerts, with a React dashboard deployed using Render and Vercel.

AI-Powered Study Assistant - React, FastAPI, Python, Ollama (Qwen3:8B), ChromaDB, RAG, PDFPlumber

Online

- Developed an AI-powered study platform that transforms uploaded PDF and PowerPoint documents into structured study notes, AI-generated quizzes, and context-aware answers using a locally hosted Qwen3:8B Large Language Model.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using document chunking, Ollama embeddings, ChromaDB vector database, and semantic retrieval to improve the accuracy and relevance of AI-generated responses.

ACHIEVEMENTS

- Patent Filed: A Wearable Alertness Monitoring Device for Detecting User Fatigue in Real Time. Application No.: 202541103516 Filed: 27-Oct-2025
- Ranked in the top 5.4% on LeetCode with 120+ coding problem submissions.